the files to be saved until the deduplication process is executed. On the flip side, this method doesn't affect the speed or performance of the storage process.

**Inline deduplication:** In this method, the deduplication is run in real-time. Thus, less storage is required. However, since the deduplication process runs as the data comes in, the speed of storage is affected, because the incoming data is checked to identify redundant copies.

## Data deduplication in Linux

Data deduplication in Linux is affordable and requires lesser hardware. The solutions are in some cases available at the block level, and are able to work only with redundant data streams of data blocks as opposed to individual files, because the logic is unable to recognise separate files over many protocols like SCSI, SAS Fibre channel and even SATA.

The file system we discuss here is Lessfs—a block level deduplication and FUSE-enabled Linux file system. FUSE is a kernel module seen on UNIX-like operating systems, which provides the ability for users to create their own file systems without touching the kernel code. In order to use these file systems, FUSE must be installed on the system. Most operating systems like Ubuntu and Fedora have the module pre-installed to support the ntfs-3g file system.

## About Lessfs and Permabit (recently acquired by Red Hat)

Lessfs is a high performance inline data deduplication file system written for Linux. It also supports LZO, QuickLZ and BZip compressions.

While Lessfs is open source, the solution provided by Permabit wasn't available until recently, when it was acquired by Red Hat. Albeiro is open source block level data deduplication software, which was launched by Permabit back in 2010, and is available as an SDK.

## Lessfs in detail

Lessfs aims to reduce disk usage where file system blocks are identical, by storing only one block and using pointers to the original block for copies. This method of storage is becoming popular in enterprise solutions for reducing disk backups and minimising virtual machine storage in particular.

It first compresses the block with LZO or QUICKLZ compression, with a combination of these methods leading to higher compression rates.

## Set up and installation

First, make sure that the requirements are all installed. These are:

- mhash
- tokyocabinet
- fuse

Go to *http://sourceforge.net/projects/mhash/files/mhash* to download the latest version of mhash. Then, download, build and install the package.

```
/*
$ tar xvzf mhash-0.X.X.X.tar.gz
$ cd mhash-0.9.9.9/
$ ./configure
$ make

$ sudo make install
*/
```

Tokyo Cabinet is the main database that Lessfs relies on. To build Tokyo Cabinet, you need to have *zlib1g-dev* and *libbz2-dev* already installed.

Download and install FUSE from *http://sourceforge.net/projects/fuse*. Now download the latest version of Lessfs from *http://sourceforge.net/projects/lessfs/files/lessfs*.

Before we start using Lessfs, we need to do some things. Go to the */etc* sub-directory inside the Lessfs source directory. Copy the Lessfs *configuration* file found there into the system's */etc* sub-directory.

```
sudo cp etc/lessfs.cfg /etc/
```

Refer to the SourceForge Lessfs page for the documentation, which is well written for any user to understand.

## Demerits

Even though Lessfs provides fast compression and data deduplication in the case of large files and small spaces, in every other case, it proves to be slow. Also, the data security it provides, even though impressive in theory, has been proven to be less effective than the solutions offered by IBM's ProtecTier or Sepaton's DeltaStor. END

### References


[1] Official Lessfs Project website: *http://www.lessfs.com*
[2] Lessfs SourceForge Project: *http://sourceforge.net/projects/lessfs*
[3] Opendedup (SDFS) Project: *http://www.opendedup.org*
[4] Wikipedia: Data Deduplication: *http://en.wikipedia.org/wiki/Data_deduplication*
[5] Notes on the Integration of Lessfs into Fedora 15: *http://fedoraproject.org/wiki/Features/LessFS*
[6] Lessfs with SCST How-To: *http://www.lessfs.com/wordpress/?page_id=577*



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